LINMA2472 Algorithms in Data Science

January 2026

Instructions

1. Write your name, surname, NOMA on each sheet.
2. Write in the frames, not outside (your exam will be scanned and only the text within the frame will appear on-screen for the grader)
3. The exam lasts 3 hours.
4. The back face (verso) of each sheet is not read, you can use it as draft paper.
5. You can use your own blank paper as draft paper.
6. Calculators and other electronic devices are forbidden.
7. You must hand in everything at the end of the exam : questions, answers, drafts.
8. Please hand in the questions in the correct order.

UCLouvain EPL Faculty January 2026	**Exam : LINMA2472** Algorithms in data science	First Name : Last Name : NOMA :

Question 1 : Kernels (applications)

Suppose that you have N datapoints $x_i \in \mathcal{X}$ with labels $y_i \in \{-1, 1\}$ that you want to classify, where $1 \leqslant i \leqslant N$. To do so, you will try two different classical methods, enhanced with the kernel trick. For the whole exercise, we introduce the following kernel-related definitions :

— $\mathcal{H}$ is an RKHS,
— $\langle \cdot, \cdot \rangle_{\mathcal{H}} : \mathcal{H} \times \mathcal{H} \to \mathbb{R}$ is the inner product associated with $\mathcal{H}$,
— $\| \cdot \|_{\mathcal{H}} : \mathcal{H} \to \mathbb{R}^+$ is the induced norm in $\mathcal{H}$ such that $\|f\|_{\mathcal{H}} = \sqrt{\langle f, f \rangle_{\mathcal{H}}}$ for any $f \in \mathcal{H}$,
— $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is the (symmetric positive-definite) kernel,
— $\phi : \mathcal{X} \to \mathcal{H}$ is a feature map such that $k(x, y) = \langle \phi(x), \phi(y) \rangle_{\mathcal{H}}$ for any $x, y \in \mathcal{X}$,
— $K \in \mathbb{R}^{N \times N}$ is the kernel matrix such that $K_{ij} = k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle_{\mathcal{H}}$.

A) Method I : hard-margin kernel support-vector machine (k-SVM). k-SVM tries to maximize the *margin* of the hyperplane which would separate the data embedded in the feature space. The decision function for any new input $x \in \mathcal{X}$ thus reads $d_I(x) = \langle v^*, \phi(x) \rangle_{\mathcal{H}} + b^*$, where the direction vector of the hyperplane $v^* \in \mathcal{H}$ and the bias $b^* \in \mathbb{R}$ are found through the k-SVM optimization problem. Let $\hat{y}_I(x)$ be the class predicted by the model x. This decision rule satisfies $\hat{y}_I(x) = \text{sign}(d_I(x))$, where the sign function is defined as $\text{sign}(t) = -1$ if $t < 0$, $\text{sign}(0) = 0$ and $\text{sign}(t) = 1$ if $t > 0$.

1. A famous theorem allows to decompose $v^* = \sum_{i=1}^{N} \phi(x_i) \gamma_i^*$, with $\gamma^* \in \mathbb{R}^N$. What is the name of this theorem ?

2. Write down the (primal) k-SVM optimization problem in terms of $\gamma \in \mathbb{R}^N$ and $b \in \mathbb{R}$, using only K, y_i and N.

3. Let $\gamma^* \in \mathbb{R}^N$ and $b^* \in \mathbb{R}$ be the solutions of the k-SVM problem (we assume that they exist and that they are bounded). Give the expression for $d_I(x)$ in terms of x, γ^*, b^*, k and x_i.

   Moreover, give the expression for $\hat{y}_I(x)$.

B) Method II : (squared) distance to the (lifted) mean. We define the centers of each class as

$$\mu_+ = \frac{1}{n_+} \sum_{\substack{i=1 \\ y_i=1}}^{N} \phi(x_i), \qquad \mu_- = \frac{1}{n_-} \sum_{\substack{i=1 \\ y_i=-1}}^{N} \phi(x_i),$$

where n_+ (resp. n_-) is the number of points labeled $+1$ (resp. -1). We call $\hat{y}_{II}(x)$ the class predicted for a new point x by the method of closest mean. The decision rule reads :

$$\hat{y}_{II}(x) = \begin{cases} 1 & \text{if} \quad \|\phi(x) - \mu_+\|_{\mathcal{H}}^2 < \|\phi(x) - \mu_-\|_{\mathcal{H}}^2, \\ 0 & \text{if} \quad \|\phi(x) - \mu_+\|_{\mathcal{H}}^2 = \|\phi(x) - \mu_-\|_{\mathcal{H}}^2, \\ -1 & \text{if} \quad \|\phi(x) - \mu_+\|_{\mathcal{H}}^2 > \|\phi(x) - \mu_-\|_{\mathcal{H}}^2. \end{cases}$$

1. Propose and motivate an expression for the decision function $d_{II}(x)$ which would satisfy the equation $\hat{y}_{II}(x) = \text{sign}(d_{II}(x))$, using only $\phi(x)$, μ_-, μ_+, $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and $\| \cdot \|_{\mathcal{H}}^2$.
   Simplify the expression as much as possible.

2. Express the squared distances $\|\phi(x) - \mu_+\|_{\mathcal{H}}^2$ and $\|\phi(x) - \mu_-\|_{\mathcal{H}}^2$ using only x, k, x_i, y_i, N, n_+ and n_-.

3. Just for this subquestion, assume that $n_+ = n_-$ (classes are balanced) and that $\|\mu_+\|_{\mathcal{H}} = \|\mu_-\|_{\mathcal{H}}$. Simplify the rule $\hat{y}_{II}(x)$ to express it only in terms of x, N, k, x_i and y_i.

   Hint : What is the relation between N, n_+ and n_- ?

C) Methods comparison.

1. Are the decision functions $d_I(x)$ and $d_{II}(x)$ affine functions of $\phi(x)$ in $\mathcal{H}$? Justify briefly.

2. If possible, give simple conditions under which both methods would be equivalent, meaning that for any new datapoint $x \in \mathcal{X}$, they would always predict the same class.

3. What method would you prefer to use for your classification task ? Choose a method and give at least two arguments in favor of it.

Question 2 : Diffusion

You would like to find a method for deblurring images. You have a program $b(x, \sigma)$ that can be used to blur an image x with blurring intensity σ. For blurring intensity $\sigma = 0$, $b(x, \sigma) = x$ and the larger σ is, the more the image is blurred.

— Given a datasets of clear images, explain how you can use the Evidence Lower Bound to train a model $f_\theta(y, \sigma)$, parameterized by θ, that attempts to predicts the clear image x that produced the blurry image $y = b(x, \sigma)$.

— You now use a trained model f_{θ^*} to deblur an image y of supposed blurred intensity σ using $\hat{x} = f_{\theta^*}(y, \sigma)$. You observe that, while this works well for small σ, the estimate is less precise for deblurring image with high blurred intensity σ. How would you suggest to improve this using an iterative approach ?

Question 3 : Implicit

Consider the function

$$f(x, \lambda) : x^3 - \lambda x^2 + x - 1$$

If $|\lambda| < \sqrt{3}$, the derivative $\partial_1 f(x, \lambda) = 3x^2 - 2\lambda x + 1$ is positive for all x. Therefore, for any $\lambda \in (-\sqrt{3}, \sqrt{3})$, there exists a unique x^* such that $f(x^*, \lambda) = 0$. Let $x^*(\lambda) : (-\sqrt{3}, \sqrt{3}) \to \mathbb{R}$ be this solution.

— We have $x^*(1) = 1$, find the value of derivative $\partial x^*(1)$.

— What is the value of the second derivative $\partial^2 x^*(1)$?

Question 4 : Hessian coloring

Consider the function

$$f : \mathbb{R}^7 \to \mathbb{R} : f(x) = \sum_{i=1}^{3} x_{2i}(x_{2i-1} - x_7) + \sum_{i=1}^{7} x_i^2$$

Our aim is to compute the Hessian $\nabla^2 f(x)$ by exploiting its sparsity.

— Find the adjacency graph of the Hessian matrix :

— How many colors would be used with **star** coloring ?

— How would the nonzero entries of the Hessian be obtained from Hessian-Vector Products using this star coloring ?

— How many colors would be used with **acyclic** coloring ?

— How would the nonzero entries of the Hessian be obtained from the Hessian-Vector Products using this acyclic coloring ?